Wide ResNets

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Background/Motivation

- Deep residual networks emphasize learning through activations within ResNet blocks while scaling performance with increased network depth.
- **Identity mapping is both a strength and weakness of residual networks.**
- There is no guarantee gradients go through to residual blocks
- Many blocks contributing little to the final goal, also known as the **diminishing feature reuse problem**.
- WRN's is an alternative approach to reduce layer count while improving model speed.

Objectives/Goal

- Replicate the results showing an ~4% error rate with WRN–28-10, showing an ~0.9% decrease in comparison to ResNet-1001.
- Demonstrating the **higher parameter model, achieves greater accuracy, with greater efficiency.**
- Holding depth constant, a wider WRN results **in lower test loss. Comparing WRN-40-1, WRN-40-2, and WRN-40-4 and demonstrating this decrease in test loss.**

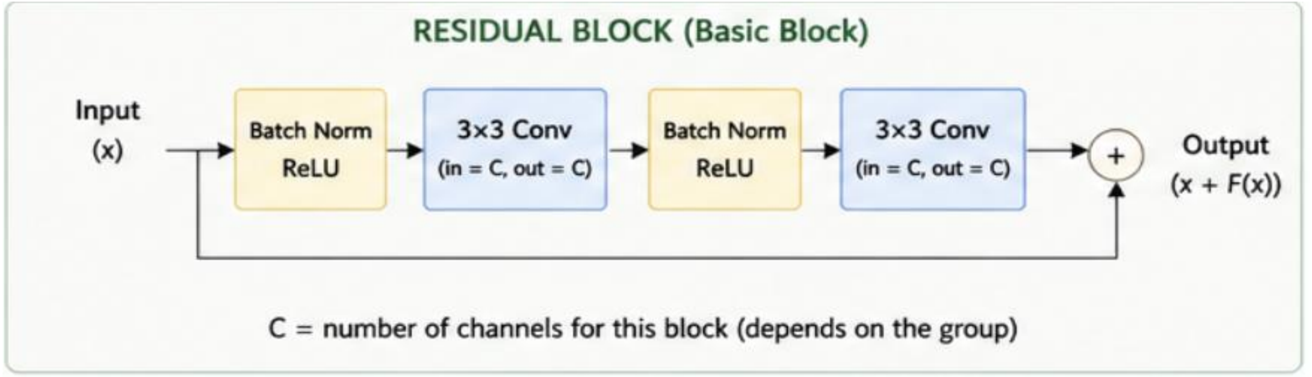
Methodology

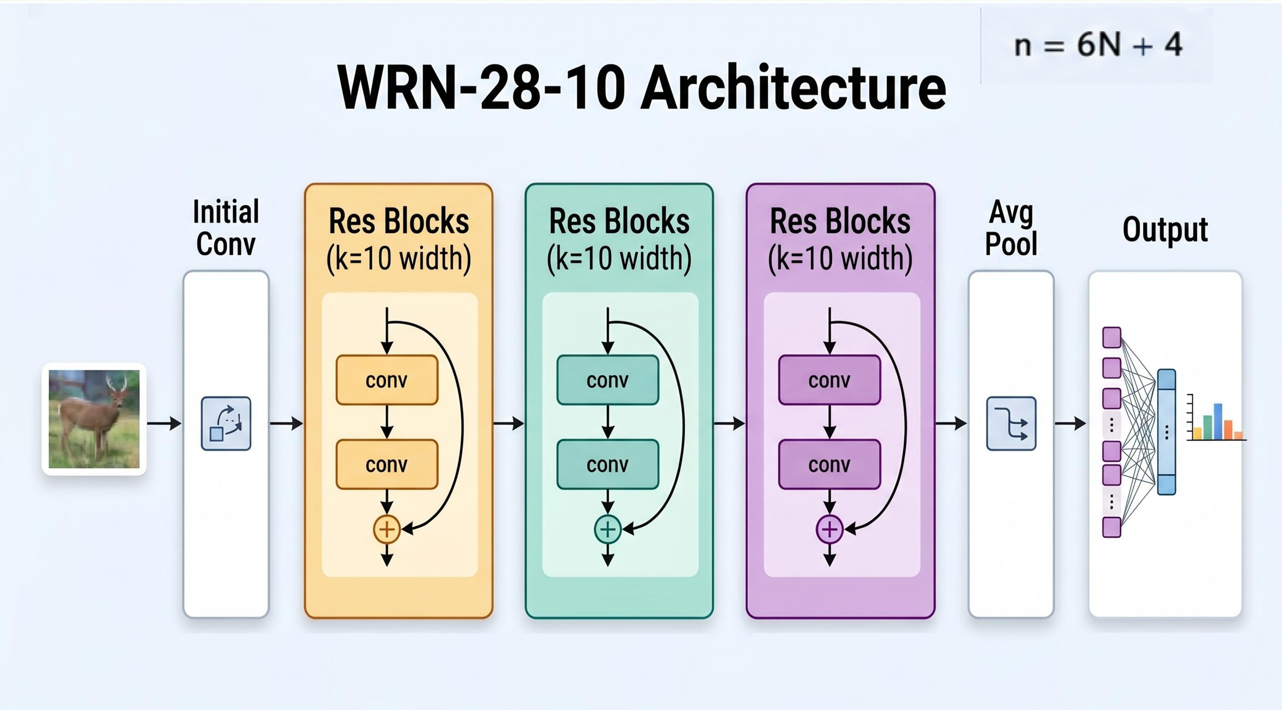
CIFAR-10

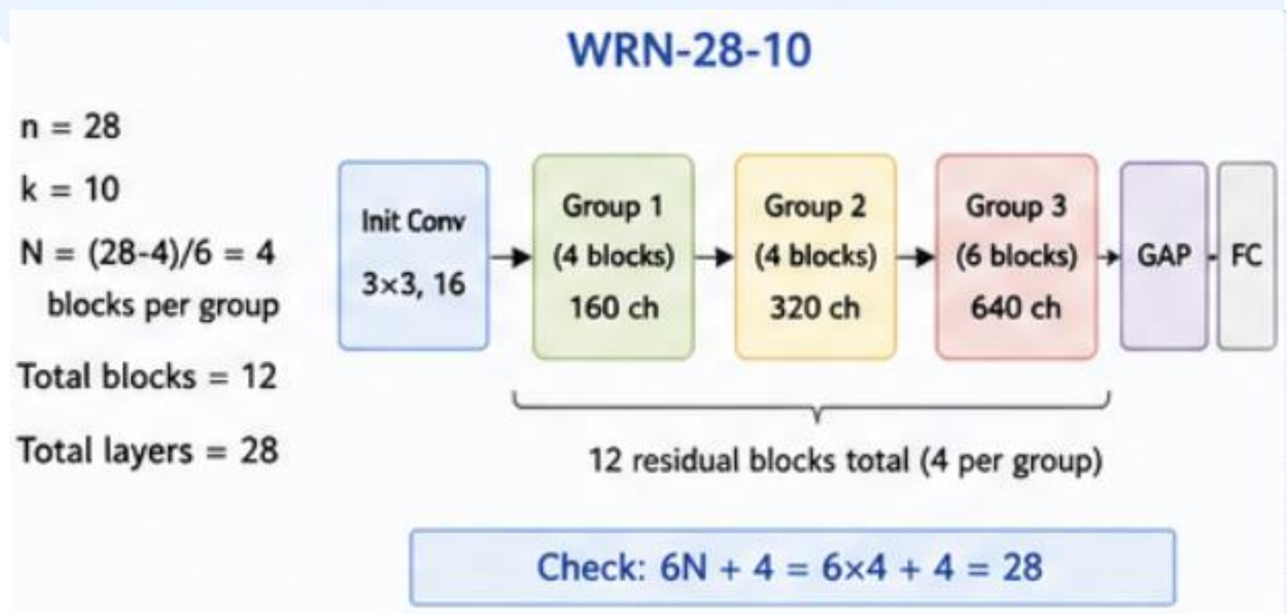
- **10 class** image dataset of 60k pictures used to train and test the WRN Model
- **Paper makes use of CIFAR-10 & CIFAR-100, as well as SVHN** to explore the impacts of dropout in this procedure.

Design

- We replicated the SGD with Nesterov momentum and cross-entropy loss.
- Initial learning rate set to **0.1**, weight decay to **0.0005**, momentum to **0.9**
- Dropping learning rate by 0.2 at **60, 120**, and **160** epochs (200 total).
- Each network consists of an initial 3×3 conv producing 16 channels, followed by three groups of (n - 4)/6 residual blocks with channel counts 16k, 32k, 64k respectively

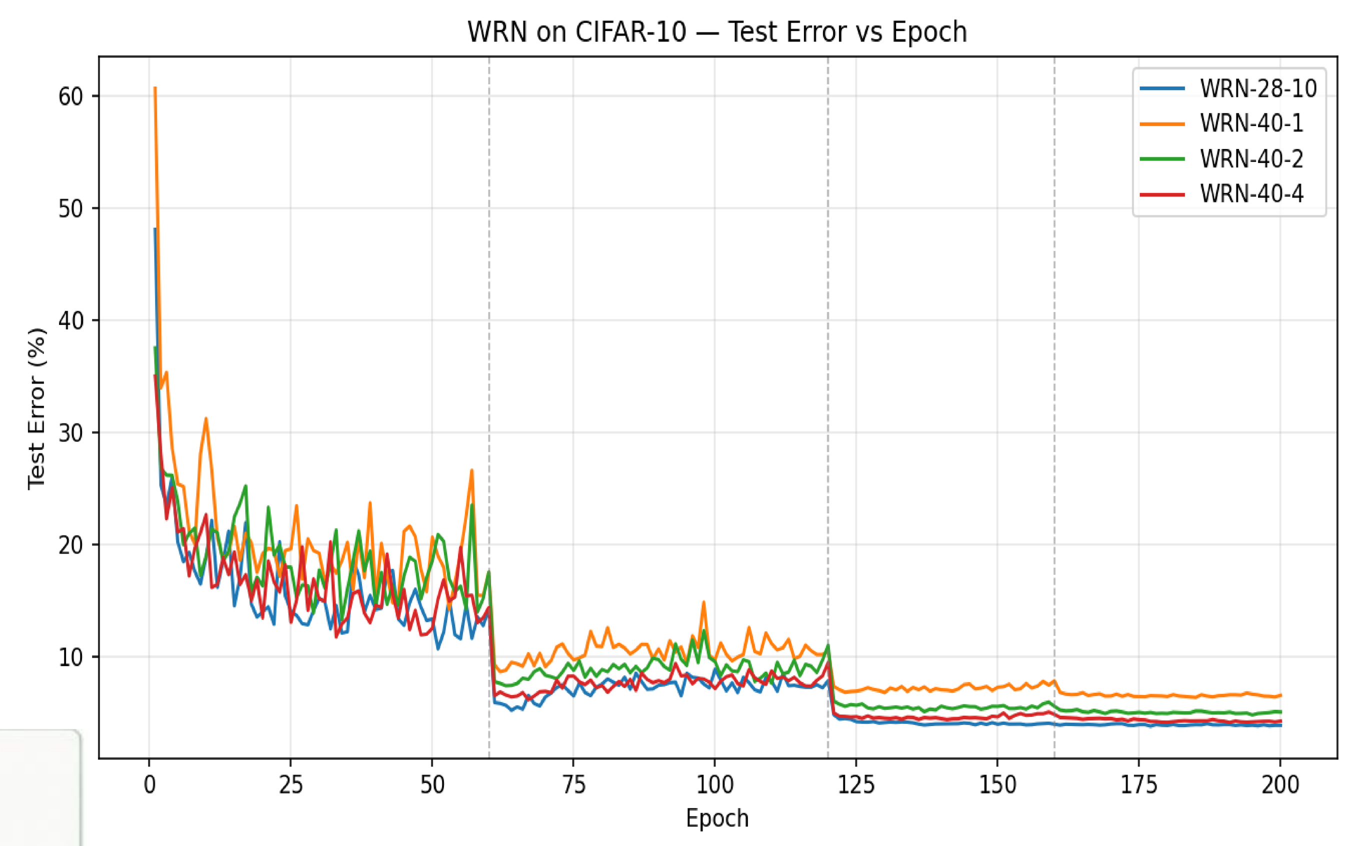






- We trained each configuration **once** rather than five times,
- We did not implement the dropout-in-residual-blocks experiments, the depth ablation, or the CIFAR-100 / SVHN / ImageNet experiments from the original paper

Results



Config	Params (M)	Our Error (%)	Paper Error (%)	Delta (%)	Total Time (min)
WRN-28-10	36.48	3.93	4.00	-0.07	38.0
WRN-40-1	0.56	6.61	6.85	-0.24	46.3
WRN-40-2	2.24	5.12	5.33	-0.21	47.7
WRN-40-4	8.95	4.32	4.97	-0.65	47.2

pre-act-ResNet[13]	110	1.7M	6.37	-
	164	1.7M	5.46	24.33
	1001	10.2M	4.92(4.64)	22.71
WRN (ours)	40-4	8.9M	4.53	21.18
	16-8	11.0M	4.27	20.43
	28-10	36.5M	**4.00**	**19.25**

- We mitigate overfitting with batch normalization, while the paper further incorporates dropout between convolutional layers as an additional regularization technique.
- These findings also align with hardware optimization benefits, as wider architectures enable more efficient parallel computation

CHANNEL DIMENSIONS BY GROUP (depending on widening factor k)

Group	WRN-*-1 (k = 1)	WRN-*-2 (k = 2)	WRN-*-4 (k = 4)	WRN-*-10 (k = 10)
Group 1	16	32	64	160
Group 2	32	64	128	320
Group 3	64	128	256	640

In general: Group 1 → 16k channels, Group 2 → 32k channels, Group 3 → 64k channels

Conclusion

- Increasing the width of residual blocks is a highly effective, computationally efficient alternative to scaling network depth.
- WRNs solve the **diminishing feature reuse problem**. By successfully replicating the paper's findings, we confirmed that a well-designed wide network can dramatically outpace deep networks in both training time and final accuracy.

Summary & Future Work

- By implementing a WideResNet-28-10 architecture and evaluating it on the CIFAR-10 dataset, we aim to show that wider networks successfully overcome the problem of diminishing feature reuse.
- This approach achieves higher classification accuracy and faster training convergence compared to deep residual networks.
- dropout layers between the convolutions in the residual blocks to prevent overfitting, a technique highly recommended for wider networks.
- Use CIFAR-100 to analyze the limits of the widening factor.

Citations

Zagoruyko, S., & Komodakis, N. (2016). Wide Residual Networks. *arXiv preprint arXiv:1605.07146*. https://arxiv.org/abs/1605.07146

Krizhevsky, A., Nair, V., & Hinton, G. (2012). CIFAR-10 (Canadian Institute for Advanced Research). http://www.cs.toronto.edu/~kriz/cifar.html